

Problem 1

1. The coordinate of point B is $(40, 40)$ and the coordinate of point C is $(20, 30)$.
2. The Income effect is -20 and the substitution effect is -10 .
3. The cross price elasticity is 0.27 .

Problem 2

1. The aggregate demand is: $Q_D = 250,000 - 8p$ if $0 \leq p \leq 25,000$ and $Q_D = 100,000 - 2p$ if $p > 25,000$.
2. The Firm's supply schedule is: $q_S = \frac{p-100}{20}$ for $p \geq 100$.
3. The equilibrium price is $p^* = 590.55$ and the equilibrium quantity is $Q^* = 245,275.6$.
4. Firms will exit.
5. The long run equilibrium price is $p^{LR} = 732.46$, the long run equilibrium quantity is $Q^{LR} = 244,140.32$ and the number of firms is 7721 .